



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E ECE B

Course Code & Title: EC8501 Digital Communication

Name of the Faculty member (s): Mr.P.Gunasekaran

Innovative Practice Description

- **Unit / Topic:** Unit IV / BER Comparison of Digital Modulation Schemes
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Jigsaw
- **Justification:**

Digital modulation is the process of encoding a digital information signal into the amplitude, phase, or frequency of the transmitted signal. The encoding process affects the bandwidth of the transmitted signal and its robustness to channel impairments. In general, a modulation technique encodes several bits into one symbol, and the rate of symbol transmission determines the bandwidth of the transmitted signal. Since the signal bandwidth is determined by the symbol rate, having a large number of bits per symbol generally yields a higher data rate for a given signal bandwidth. However, the larger the number of bits per symbol, the greater the required received SNR for a given target BER.

Jigsaw is a collaborative group activity with a twist: students effectively teach each other (with the teacher's guidance). Students learn through the process of communicating with one another about a given skill or procedure, topic or problem. This topic and its related to modulation schemes are more suitable for a Jig-Saw cooperative learning activity in which students get exposure in different digital modulation schemes. Students should be able to enhance their communication, peer learning, individual and teamwork, and self-learning by this activity

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

The heterogeneous groups with six members were formed. Each group formed by bright students, average students, and slow learners, consists of 4 members. One group contains 5 members. Classification of students such as a bright student, average students, and slow learners is based on their academic performance. The total class strength is 41 (Boys and Girls). Seven heterogeneous groups were formed based on academic performances such as the CGPA of end semester examination. The table1 shows the name of the groups.



Sl.No	Group Name	Number of members
1	Mercury	6
2	Venus	6
3	Mars	6
4	Jupiter	6
5	Saturn	6
6	Neptune	6
7	Pluto	5

Sub Topics:

- BER of Binary Phase Shift Keying
- BER of Binary Frequency Shift Keying
- BER of Quadrature Phase Shift Keying
- BER of Quadrature Amplitude Modulation
- BER of Differential Phase Shift Keying
- BER of M-ary Phase Shift Keying
- Comparison of BER among all digital Modulation Schemes

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1	PSO3
CO2	3	3	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PSO1
	3	2
Justification for correlation	The outcome is mapped to level 3 because Modulation and demodulation concepts are essential to understand the digital modulation techniques.	The students will be able to understand the band pass transmission schemes for designing the special antennas and the underlying circuitry in the area of communication, hence it is highly mapped.



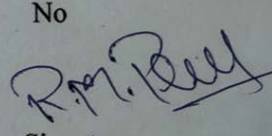
- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

It piqued the curiosity of the students. They learn how to work as a team and communicate with one another.

<u>FEEDBACK FORM</u>	
Innovative Practice: Jigsaw	
Is encourage cooperative learning among yourself?	☒ Yes ☐ No
Connection with course material in a creative and engaging way?	☒ Yes ☐ No
Is improve listening, Communication and problem solving skills?	☒ Yes ☐ No
 Signature	

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Every student had an equal opportunity to participate in this activity. The success of the exercise was assessed by asking the same question in Internal Assessment Test II, which was answered correctly by around 70% of students.



❖ *Challenges faced in implementation:*

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ B.Sklar, “Digital Communication Fundamentals and Applications”, 2nd Edition, Pearson Education, 2009.
- ❖ B.P.Lathi, “Modern Digital and Analog Communication Systems” 3rd Edition, Oxford University Press 2007.
- ❖ <https://www.gaussianwaves.com/2010/04/performance-comparison-of-digital-modulation-techniques-2/>
- ❖ <https://www.jigsaw.org/>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

Signature of Faculty Member

HOD



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Course Code & Title: EC8501 Digital Communication

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Innovative Practice Description

- **Unit / Topic:** Unit V / Channel Coding Theorem
- **Course Outcome:** CO 5 & CO 6
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Flipped Classroom
- **Justification:**

The purpose of channel coding theory is to find codes which transmit quickly, contain many valid code words and can correct or at least detect many errors. While not mutually exclusive, performance in these areas is a trade-off. So, different codes are optimal for different applications. The objective of collaborative learning is to enhance communication, creativity and problem-solving skills. This domain helps the students to get placement in core fields.

A sequence of evaluations is used in a collaborative learning strategy to help pupils master a subject or topic. It focuses on successful group interaction among students in order for them to grasp certain subjects. Additionally, this teaching method improves students' communication and problem-solving abilities. Because the textbook only has five pages, it does not receive additional attention in parts of the syllabus and test. As a result, I consider this a matter for independent research. Because all students value independence and want to learn on their own.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

A six-person heterogeneous group was created. There were clever pupils, ordinary students, and slow learners in each group. Academic performance determines whether a pupil is classified as a brilliant student, an average student, or a slow learner. The total number of students in the class is 41. (Boys and Girls).

Specific topic was given to the specific group to learn on their own. Resources like reference book, videos were given to the student group. Students are asked to prepare more in depth than before. Get the class back together to share the individual group's work with everyone.



Sub Topics:

- Introduction to Channel Coding theorem
- Linear Block Codes
- Hamming Code
- Syndrome Calculation of Hamming Code
- Cyclic Codes
- Convolutional Codes
- Viterbi Decoding

• CO – PO / PSO mapping:

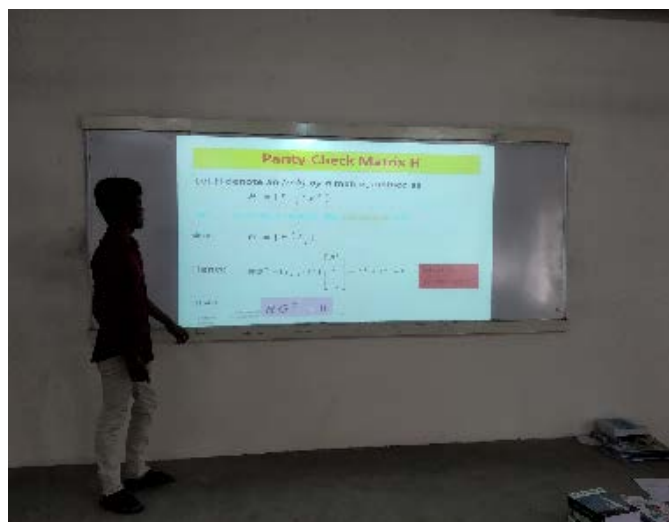
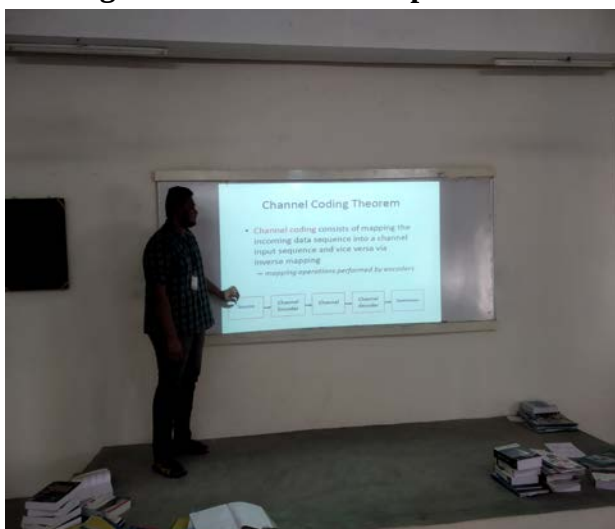
CO	PO1	PO2	PO3	PO4	PSO1	PSO3
CO2	3	3	2	1	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO3	PSO3
	2	2
Justification for correlation	The students will be able to develop coding methods are to bring good solutions for complex problems and to reduce errors; hence it is mapped to level 2.	The students will be able to design the channel encoder and decoder circuit in the communication system; hence it is mapped to level 2.

• Images / Screenshot of the practice:





- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

The feedback was collected from the students using the following questionnaires. It is used to assess the success of the Flipped Classroom. The feedback was collected from the students using the following questionnaires. It is used to assess the success of the Flipped Classroom. The first question is used to assess the student's willingness whether the students are like this activity or not. The second and third questions measure the objectives and outcomes of this activity. The fourth question asks the students comments to improve the learning strategies.

Sample Feedback Form:

FEEDBACK

1. Have you liked Flipped Classroom learning activity?
 - a. Yes
 - b. No
2. Whether the Flipped classroom gives me greater opportunities to enhance my interpersonal skills like communication, listening, team work and leadership?
 - a. Strongly Agree
 - b. Agree
 - c. Disagree
3. Whether Flipped Classroom will help me in my future education?
 - a. Yes
 - b. No
4. Any other comments



Sample Feedback from students:

FEEDBACK FORM

1. Have you liked Flipped Classroom learning activity?
a. Yes ☒
b. No ☐
2. Whether the Flipped classroom gives me greater opportunities to enhance my interpersonal skills like communication, listening, team work and leadership?
a. Strongly Agree ☒
b. Agree ☐
c. Disagree ☐
3. Whether Flipped Classroom will help me in my future education?
a. Yes ☐
b. No ☐
4. Any other comments
Enjoyed the flipped classroom activity.

Signature _____

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- It promotes peer interaction among students and collaboration skills
- It encourages bright students' engagement.
- It provides increased individual attention.
- It makes independent learning rather than teaching

Every student had an equal opportunity to participate in this activity. The activity's success was assessed by asking the same question in Internal Assessment Test III – Around 75% of students correctly answered.

❖ **Challenges faced in implementation:**

The fundamental issue is that few students have been exposed to flipped classrooms. Students struggle with self-discipline and may show up to class without having fully comprehended the material. To encourage all students to participate, I began by praising others, taking turns for equitable involvement, and allowing students to make decisions together.



References:

- ❖ B.Sklar, “Digital Communication Fundamentals and Applications”, 2nd Edition, Pearson Education, 2009.
- ❖ B.P.Lathi, “Modern Digital and Analog Communication Systems” 3rd Edition, Oxford University Press 2007.
- ❖ Stephanie butler velegol, sarah e. Zappe, emily mahoney, The Evolution of a Flipped Classroom: Evidence-Based Recommendations, winter 2015, pp.1-37.
- ❖ Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™

Signature of Faculty Member

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